# ASSIGNMENT 1 – CO1 | TASK 5

## Expressions and Operator Precedence

**Scenario:** Namma Metro Smart-Card Fare System

### Objective / explanation

Multiplication has higher precedence than addition. Post-increment returns the old value; pre-increment changes the value before it is used. The peak-hour expression uses half-open intervals.

### C++ source code

```
#include &lt;iostream&gt;
using namespace std;

int main() {
    int x = 5, y = 3;
    cout &lt;&lt; x + y * 2 &lt;&lt; ' ' &lt;&lt; (x + y) * 2 &lt;&lt; '\n';
    int z = x++ + ++y;
    cout &lt;&lt; z &lt;&lt; ' ' &lt;&lt; x &lt;&lt; ' ' &lt;&lt; y &lt;&lt; '\n';
    cout &lt;&lt; (x &gt; y ? x : y) &lt;&lt; ' ' &lt;&lt; x % 4 * 2 &lt;&lt; '\n';
    cout &lt;&lt; 10 / 3 &lt;&lt; ' ' &lt;&lt; 10 % 3 &lt;&lt; ' '
         &lt;&lt; -10 / 3 &lt;&lt; ' ' &lt;&lt; -10 % 3 &lt;&lt; '\n';
    int k = 0;
    for (int i = 0; i &lt; 10; i += 3) k += i;
    cout &lt;&lt; k &lt;&lt; '\n';
}
```

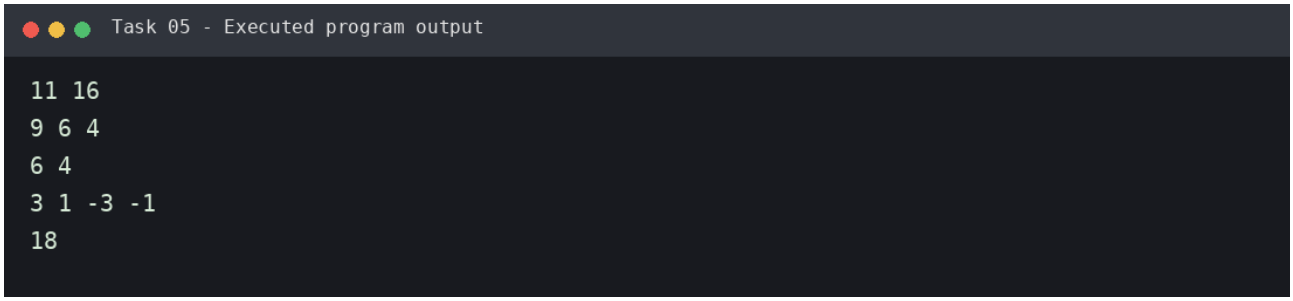
### Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

### Actual captured program output

```
11 16
9 6 4
6 4
3 1 -3 -1
18
```

### Output screenshot




### Analysis / complexity

Peak check: (hour >= 8 && hour < 10) || (hour >= 17 && hour < 20). Exactly 10:00 and 20:00 are off-peak. Integer division truncates toward zero.

**Evidence note:** Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.